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### CONTRIBUTION

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**Title:** Proposal for update in baseline text of draft Recommendation ITU-T L.CArural  
"Low-cost sustainable ICT solutions for climate change adaptation in rural areas  
and communities"

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**Abstract:** In this contribution, the draft Recommendation ITU-T L.CArural has been improved by adding new section and content under existing section. It includes addition of new section on 'Challenges and Way forward' and detailing under heading of 'Proposed Technologies', 'Adaptation and Sustainability Activities', 'Recommendations and future use',

## **Draft Recommendation ITU-T L.CArural**

### **"Low-cost sustainable ICT solutions for climate change adaptation in rural areas and communities"**

#### Summary

This Recommendation will provide a set of low-cost and sustainable ICT solutions tailored to the context of rural areas and communities, including climate monitoring, building climate resilient in rural agriculture systems and many more. The objective of this Recommendation is to enhance the ability of rural areas and communities to respond to climate risks and expand their institutional capacity to accelerate climate adaptation actions.

The livelihood of rural areas and communities often rely on climate sensitive resources, such as agricultural resources and water supply. Climate change, from higher temperature to drought and heavily rainfall, can directly impact the availability of these resources. At the same time, rural areas and communities are also lacking in means to adopting climate change adaptation measures largely due to costs involved. The Recommendation will provide universal ICT solutions that are applicable to rural communities focussing on the Agricultural sector.

#### Keywords

## **Introduction**

ICTs can play a crucial role in supporting communities and systems to adapt to the effects of climate change. The effects of climate change often disproportionately impact vulnerable communities. These areas often lack the social and economic resources to enhance climate resiliency. This leads to a series of challenges that are barring rural communities from taking advantage of digital technologies in adapting to the effects of climate change.

In addition, the lack of adequate broadband infrastructure is also limiting the adoption of ICTs in rural areas. Low-cost, portable and energy efficient powering units and broadband infrastructures can accelerate the adoption of ICTs, improving the usage of ICT for alert system in case of emergency due to climate events and thereby enhancing their adaptation measures.

## TECHNOLOGIES. ADAPTATION AND MITIGATION

### 1. Adaptation of Present Technologies Applied in Agriculture to Climate Change

There are several technologies applied in modern agriculture that has enhanced productivity.

| No | Technology                                                                             | Application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Synnefa                                                                                | Synnefa offers smart and comprehensive farming tools with a variety of products to help the modern farmer take the guesswork out of farming. (No technology specifically identified)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2  | A smart and sustainable digital future for European agriculture and rural areas (2019) | <p>Digital technologies such as artificial intelligence (AI), robotics, blockchain, high performance computing (HPC), Internet of Things (IoT) and 5G have the potential to increase farm efficiency and improve production, and also to contribute to making farming systems more sustainable from an economic, social and environmental point of view, as it is the case in other sectors.</p> <p>Such technologies can optimise all types of farming, enable better decision making, and reshape the functioning of agri-food markets. Increased use of digital technologies will also have a positive impact on the quality of life for farm workers and the rural population, and may attract a younger generation to farming and rural business start-ups.</p> <p>However, the use of digital technology in agriculture and rural areas in the EU is, on average, low. The lack of information about existing technologies, the lack of digital skills and the limited availability of reliable cost/benefit analyses of the new technologies are challenges to increased investments in digital applications. In many cases, research and innovation are still required to develop new solutions. In other cases, the basic infrastructure, such as broadband or access to other high-speed internet connections, is partly missing.</p> |
| 3  | Rural regions of the future: Seizing technological change                              | This proposal outlines the importance of digitalisation for rural development and the well-being of rural citizens and policy responses needed to make the most of future opportunities that may emerge. It describes the trends and impacts of digitalisation in                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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|   |                                                                     | rural communities and examines the policies needed to harness the benefits brought by technological change. It maps a number of disruptive technologies and how governments may seize their benefits for rural regions. The last section outlines how the emerging opportunities for rural development can be used as a tool to reach countrywide and global Sustainable Development Goals in the 2030 Agenda.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4 | Farm force                                                          | <p>Climate Change mitigation is increasingly becoming a top requirement in the agricultural sector. Fair Farm uses Farmforce to monitor and store all farm-level data in one place hence sharing certification with auditors with ease.</p> <p>Farmforce technology is resolving the production problems associated with the most appetising spices for consumers.</p> <p>Reducing transaction costs in contract farming arrangements: the case of farmforce.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5 | FAO-UN Briefing Paper 2019                                          | <p><b>Digital Technologies in Agriculture and Rural Areas.</b></p> <p>Digital innovations and technologies may be part of the solution. The so-called 'Fourth Industrial Revolution' (Industry 4.0)<sup>1</sup> is seeing several sectors rapidly transformed by 'disruptive' digital technologies such as Blockchain, Internet of Things, Artificial Intelligence and Immerse Reality. In the agriculture and food sector, the spread of mobile technologies, remote-sensing services and distributed computing are already improving smallholders' access to information, inputs, market, 1 The term Industry 4.0 originated in Germany where it was applied to rapid transformations in the design, manufacture, operation and service of manufacturing systems and products (European Parliament, 2015a) finance and training. Digital technologies are creating new opportunities to integrate smallholders in a digitally driven agrifood system (USAID, 2018).</p> |
| 6 |                                                                     | Mobile phone applications for weather and climate information for smallholder farmer decision making. It discusses the development and testing of two proof-of-concept mobile phone applications that use elements of the PICSA approach to provide climate information and decision-making tools. The concept provides brief explanation of the PICSA approach before summarising the findings of a review that was undertaken to highlight the lessons learned from similar mobile-agriculture (mAgri) initiatives.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 7 | Digital Technologies for Agricultural and Rural Developments in the | Introducing a technology stewardship model to encourage ICT adoption in agricultural communities of practice: reflections on a Canada/Sri Lanka partnership project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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|   | global South(2018)                                                  | This report puts forward 'technology stewardship' as an approach for training and supporting individuals and teams who engage their agricultural communities to encourage and support innovative practices with low-cost, widely available digital technologies. It draws upon theoretical ideas of 'communities of practice' to analyse a stewardship initiative that has been implemented in Sri Lanka, presenting preliminary results of an ongoing action research project involving pilot studies of technology stewardship with partner organizations. It outlines plans for a Joint Education and Training Initiative for technology stewardship that includes community engagement, rapid prototyping, evaluation and impact assessment. |
| 8 | The effect of ICTs on agricultural distribution channels in Mexico. | <p>This applies the concepts of transaction cost economics and social embeddedness to examine the role of information and communication technologies (ICTs) in distribution channels for agricultural produce in Mexico.</p> <p>ICTs are observed to be central to marketing, establishing new links between farmers, markets and end consumers, and optimizing distribution channels that potentially reduce transaction costs.</p> <p>ICTs are primarily used by farming communities as a management tool for intermediation and cooperation by building relationships, and through acquisition of knowledge.</p>                                                                                                                              |
| 9 |                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## 2. Adaptation of technologies to climate change

These include;

The various technologies applied and adopted to the agriculture sector and the rural community have been disjointedly applied in different jurisdictions. This recommendation consolidates a generic identification framework for deployment of low-cost sustainable digital technologies to ensure adaptability to the effects to climate change. This ITU-T L.CArural recommendation will provide global references in the immediate and future innovation and applications.

## 3. Low-cost technologies in agriculture

The different technologies can be adapted to sustainability by lowering their deployment costs in the rural communities. The target and focus on the agriculture sector herein provide opportunity to deliver gains made in a more structured and widely adopted approach for immediate use in the rural communities.

#### 4. Proposed Technologies

The technologies can be singularly or jointly deployed to attain a seamless sustainable system for rural communities and agriculture. The aim is to deploy Smart Sustainable Application to various sectors in rural communities and agriculture such as fisheries, crop farming, livestock... and related Infrastructure.

**Commented [KD51]:** This table to talk about the applications of technology and not describe the technology as this is done in SRDT

| No | Technology                    | Propose Usage and Applications                                                                                                                                                                                                                                                                       | Requisite adaptation                                                                            |
|----|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 1  | Artificial intelligence (AI), | Analysis of data by AI helps to know with precision the extend of the agriculture exploitation to detect weeds and disease on the Soil culture and inform the quantity of fertilizer to be used. (soil sample testing before planting.)                                                              | Soil testing and preseason works                                                                |
|    |                               | On irrigation, AI coupled with sensors makes it possible to manage the water supply automatically in the drier part of the plot/farm and hence reduce overall water needs.                                                                                                                           | Alternative power supply e.g., Solar                                                            |
| 2  | Robotics ()                   | WITTRA (Out of Box Wireless sensor) used along the IoT as an application for monitoring objects and assets in any environment. Used for soil management, crop monitoring, local or zonal temperature and humidity, preventive maintenance, livestock monitoring and management automation. (Sweden). | Adopt. Since robotics can outdo human resilience. Part of smart farming by use of technologies. |
| 3  | Blockchain                    | Block chain can deliver transference to complex global food supply chains and has potential to guide current food production towards better sustainability and efficiency.                                                                                                                           | June 2022 paper by MDBI.....                                                                    |
|    |                               | Can store and share data e.g. on Humidity and temperature during transportation and storage of food to prevent food-borne diseases, spoilage and household food waste.                                                                                                                               |                                                                                                 |
| 4  | High Performance              | High performance computing (HPC) is the ability to process data and                                                                                                                                                                                                                                  | Establishment of data centres to consolidate applications that rural                            |

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|   | Computing (HPC),         | perform complex calculations at high speeds. To put it into perspective, a laptop or desktop with a 3 GHz processor can perform around 3 billion calculations per second. While that is much faster than any human can achieve, it pales in comparison to HPC solutions that can perform quadrillions of calculations per second                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | feeds can be done and enhance speed of production.<br><br>It is through data that ground breaking scientific discoveries are made, game-changing innovations are fueled, and quality of life is improved for billions of people around the globe. HPC is the foundation for scientific, industrial, and societal advancements. |
| 5 | Internet of Things (IoT) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                |
| 6 | 5G technology            | <u>5G technology will provide high speed internet services (eMBB), ultra-reliable Low Latency Communication uRLLC (latency &lt;1ms) and Massive MTC (large number of devices may be connected per square km). Once the 5G technology is deployed in the country and the rural areas are covered then high speed connectivity may be used for various applications. 5G technology in precision agriculture will ensure greater profitability and efficient utilization of resources by use of automated tractors / harvesters, precision seeders, and automated weed &amp; pest controllers. Drones with 5G technology may be used efficient &amp; precise spraying of fertilizers in fields, and also to scan and identify unwanted weeds through the use of AI. It will help farmers to better organize and allocate their time and attention towards areas that really need it.</u><br><br><u>Massive M2M and uRLLC features of 5G may also be used for remote monitoring and maintenance of machines centrally and that data could be transmitted in real-time.</u> | <ul style="list-style-type: none"> <li>• <u>e-Governance</u></li> <li>• <u>Remote surgery</u></li> <li>• <u>Drones,</u></li> <li>• <u>Remote maintenance of machines</u></li> <li>• <u>Precision agriculture</u></li> <li>• <u>Livestock monitoring and management</u></li> </ul>                                              |
| 7 | Farm force,              | (to be explored for proper study and                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | To be developed further                                                                                                                                                                                                                                                                                                        |



|    |                         |                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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|    | mAgri..                 | application)<br>Is a mobile service that links small holder farmers to other actors in the agro-value chain...                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 8  | Immerse Reality         |                                                                                                                                                                                                                                                                                                                                                                                                               | To be developed further                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 9  | Remote-sensing services | <a href="#">Remote sensors are used to capture and store the geographic information, further analyse, manipulate, manage and present all types of spatial or geographical data. These sensors are used in various agriculture applications such as crop assessment, forecasting yield dates, yield modelling and forecasting, identification of plants and pests, land cover and degradation mapping etc.</a> | <a href="#">Some of the common sensors of smartphone may have common agriculture usage: -</a><br><ol style="list-style-type: none"> <li><a href="#">1. Image Sensors (Camera):- may be used in Disease detection, chlorophyll status, Fruit ripeness, Leaf area index (LAI), Harvest readiness, Soil erosion and other analysis.</a></li> <li><a href="#">2. Global Positioning System (GPS):- used for machine driving &amp; control, and tracking, land management and crop mapping.</a></li> <li><a href="#">3. Barometer may be used to measure the elevation height in hilly agriculture.</a></li> <li><a href="#">4. Microphone may be used for machine maintenance , Bug detection to make audio queries.</a></li> </ol> |
| 10 | Distributed computing   |                                                                                                                                                                                                                                                                                                                                                                                                               | To be developed further                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 11 | Technology Stewardship  | This is a concept of bringing people together to use technology for adaptation to climate change challenges.                                                                                                                                                                                                                                                                                                  | To be developed further                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## 5. Adaptation and Sustainability Activities

[Adaptation and sustainability activities in rural areas can help to improve the resilience of rural communities to the impacts of climate change and promote sustainable development. Here are some examples of such activities:](#)

- [1. Agroforestry: Planting trees on farms and in communities can help to restore degraded land, provide shade for crops, and protect against soil erosion and desertification. Agroforestry can also promote biodiversity, improve soil fertility, and provide additional sources of income for farmers through the sale of timber, fruits, and other forest products. The role of surveillance in Agroforestry applications is highly critical, due to the need to](#)

cover large areas. Unmanned Aerial Vehicles (UAVs) with sensors and camera may enable surveillance of the crop field. The focus may be placed on enabling smart village with integrated agriculture, comprising animal husbandry and fisheries, in addition to traditional farming. Agroforestry will also be adopted in these villages.

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**Water conservation and management:** Adopting water conservation and management practices, such as rainwater harvesting, smart irrigation, drip irrigation, and water recycling, can help to reduce water waste, increase crop yields, and improve water quality. This is particularly important in areas that are prone to drought or water scarcity.

2. **Sustainable land management:** Implementing sustainable land management practices, such as minimum tillage or no-till farming, cover cropping, and crop rotation, can help to conserve soil moisture, reduce erosion, and increase soil fertility. This can improve crop yields, reduce greenhouse gas emissions, and promote the long-term sustainability of agricultural production systems.
3. **Renewable energy:** Investing in renewable energy technologies, such as solar, wind, or biogas, can help to reduce reliance on fossil fuels and promote sustainable energy practices in rural communities. Renewable energy can also provide access to electricity in areas that are off-grid, improving rural livelihoods and enabling the use of modern technologies.
4. **Sustainable forest management:** Adopting sustainable forest management practices, such as reduced impact logging, forest restoration, and community-based forest management, can help to promote the conservation of forest ecosystems and biodiversity, while also providing sustainable sources of timber and other forest products.
5. **Climate-smart agriculture:** Implementing climate-smart agriculture practices, such as crop diversification, improved seed varieties, and integrated pest management, can help to increase agricultural productivity and resilience while reducing greenhouse gas emissions and other negative environmental impacts.

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## 6. Best Practice (case studies)

| No | Sector                         | Application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 1  | Food Security and Water Supply | <b>Scenario</b><br>Climate change endangers the quality and availability of water and food. Currently a large population of the world is experiencing increased water stress, and crop yields in some countries are expected to drop by half.. The first step to address food security is to systematically monitor world food supplies, including the mapping of agricultural production and food shortages. Monitoring environmental and soil conditions using ICTs can make farming more profitable and sustainable. Better water management using ICTs can improve the overall efficiency of water use, providing significant savings and a more sustainable use of water resources. |

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|   |                                    | <p><b>Application of Resilient Digital Technologies</b></p> <p>The deployment of ICTs will enable farmers to better-forecast crop yields and production. This data can then be shared to increase the number of farmers profiting from the information.</p> <p>ICTs that can be used include:</p> <ol style="list-style-type: none"> <li>1. ICT tools used in agricultural and soil monitoring include sensors and telemetry units that measure and transmit parameters such as air temperature, humidity, leaf wetness and soil moisture over mobile networks to global databases</li> <li>2. Machine-to-machine (M2M) connectivity that supports remote sensing infrastructure, with high-resolution radiometers and moderate-resolution imaging spectrometers used to monitor food and water resources.</li> <li>3. PCs, mobile devices, servers, mainframes and network databases used for food security analysis, modelling and mapping.</li> <li>4. Communications infrastructure, including the Internet, to distribute information to farmers and consumers.</li> </ol>                                                                                                                                                                                                                    |
| 2 | Deforestation and Waste Management | <p><b>Scenario</b></p> <p><b>Deforestation is a phenomenon that has contributed heavily to the negative climate change. The proliferation of electronic waste has made it worse as the demand for newer products push obsolescence of current technologies at alarming rates.</b></p> <p><b>Application of Resilient Digital Technologies</b></p> <ol style="list-style-type: none"> <li>1. ICTs can contribute to this issue, for instance, by developing technological paths to sustainability and protection of tropical forest, as well as to enhance data collection on the condition of the forests.</li> <li>2. Satellites that are now able to take images through clouds and at night and remote sensing applications are critical for monitoring the health of the world's tropical forests trees and deforestation of these vast forests.</li> <li>3. ICTs can also play a role in environmental protection; waste management and environmentally friendly supply chain management.</li> <li>4. It is now possible to search websites for places to recycle ICT equipment, including mobile phones.</li> <li>5. ITU, together with the ICT industry, is developing standards for improved recycling of industrial waste, including a recommendation on communication formats</li> </ol> |
| 3 | Health Care and Medical Services   | <p><b>Scenario</b></p> <p>The world is encountering unprecedented changes in the atmosphere and depletion of soil fertility, aquifers, ocean fisheries and biodiversity in general. It is recognized that such changes will affect economic activities and infrastructure and poses risks to human health<sup>38</sup>. Seasonal death rates due to unexpected extremes in</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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|   |                              | <p>temperature are expected to increase and the seasonal pattern of vector borne diseases will change. Standard observational epidemiological methods can illuminate the health consequences of local climatic trends by comparison of data-sets. ICT enhances our capacity to process and share data and subsequently to estimate future impacts.</p> <p><b>Application of Resilient Digital Technologies</b></p> <p>ICT improves self-help online education and information dissemination. For those who have internet access it is simple to obtain information about what to do to minimize health and climate risks using online encyclopedias, and then to find more detailed information through search engines.</p> <p>Use of ICTs in healthcare is known as eHealth (or Health informatics). Healthcare practice supported by electronic processes and communication.</p> <p>ICT tools used in eHealth include not only computers but also clinical guidelines, formal medical terminologies, and information and communication systems. It is applied to the areas of nursing, clinical care, dentistry, pharmacy, public health and (bio) medical research.</p> <p>Remote health monitoring enables individuals to continue to lead independent lives in their own homes. Telemedicine can also provide patients with access to specialists outside their geographic area, using a broadband network. This removes the need for the patient to travel to the doctor's office and so reduces GHG emissions.</p> |
| 4 | Transport Infrastructure and | <p><b>Use of Smart Logistics</b></p> <p>The potential systemic impact of ICTs is particularly apparent in the transport sector which, according to the Stern report, accounts for 14 per cent of total GHG emissions. Through a host of efficiencies in transport and storage, smart logistics in Europe could deliver fuel, electricity and heating savings of 225 MtCO<sub>2</sub> e. The global emissions savings from smart logistics in 2020 was estimated to reach 1.52 GtCO<sub>2</sub> e, with energy savings worth USD 441.7 bn1. Although the main focus of Intelligent Transport Systems (ITS) is on the safety, management and efficiency of transport, ITS can also be used to reduce their environmental impact. For example:</p> <ol style="list-style-type: none"> <li>1. Using GPS for navigation or vehicle dispatch can reduce journey times. Drivers with navigation saw their fuel efficiency increase 12 per cent, with fuel consumption falling from 8.3 to 7.3 l/100kms. This increase in fuel economy translates to a 91 tonnes decrease in carbon dioxide emissions every year per driver, or 24 per cent decrease over the amount that the average non-navigation user emits per year<sup>46</sup>.</li> <li>2. Using the "always turn right rule", or using a mobile phone (or PDA) to inform the driver of the next destination.</li> </ol>                                                                                                                                                  |

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|   |                      | <ol style="list-style-type: none"> <li>3. Smart traffic control, where traffic lights send out status signals to warn drivers if they need to slow down or stop.</li> <li>4. Smart parking, where vehicles are directed to an empty space so that there is no need to drive around to find a slot. This reduces engine-running time.</li> <li>5. Road pricing schemes, such as the congestion charge in London, can encourage greater use of public transport and reduce congestion, thereby reducing journey times.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5 | Energy Efficiency    | <p><b>Scenario</b><br/>ICTs can be used to maximize the efficiency of power systems. Their computing and communications capabilities are essential if power from renewable resources such as geothermal, solar, wind, wave and tidal effects are to be harnessed efficiently and fed into the electricity grid in a smart way. ICTs are required to control the load on the grid by maximizing the utilisation of available solar, wind and tidal power</p> <p>ICTs are able to model the real-time status of renewable energy systems taking into account local weather stations, so that transmission losses are minimized by selecting the shortest route from source to load.</p> <p><b>Resilient Applications</b><br/>The efficiency of power and cooling systems in ICT networks, support development of energy efficiency ICT architectures/design.<br/>Added energy saving features to ICT equipment and applications,<br/>Improved airflow controlling technology, cooling technology and renewable energy systems and more. These features can improve energy efficiency and reduce carbon emissions of digital technologies.</p> |
| 6 | Telecommunications   | <p>The need for travel can be reduced by using <b>virtual meetings</b> accessible to all users. The most common are web-based conferencing services that require Internet access and web-based software, allowing virtual meetings from different locations, including the sharing and exchanging of documents. Other services include teleconferencing that allows for multiple participants in one phone call, and videoconferencing with both audio and video transmission of meeting activities. Both of these can replace or complement face-to-face meetings. Telepresence, used especially by big companies and governmental organizations, provides high-definition video, life-sized images, spatial audio, imperceptible latency and easy set up and use. This requires one or more display screens with microphones, speakers and cameras specially designed for the telepresence system.</p> <p>Ordinary work can also be done remotely instead of requirement to have all participants in the seated build environment. This is widely adopted especially after the Covid 19 pandemic.</p>                                     |
| 7 | Fisheries and Marine | The article on the role of ICTs in fisheries by the University of                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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|   |                 | <p>Nebraska Lincoln (2016) identifies the benefits to farmers using ICTs.</p> <p>“The roles of ICTs are numerous and important in fish farming and production. It is shown that knowledge sharing among fish farmers and monitoring of illegal fishing are the major roles of ICTs in fish farming as indicated by 100% response. Other major roles were information exchange and emergency alerts (95.3%), sales and marketing (91.3%), prices information provisions (98.6%), food &amp; livestock security issues (83.3%), increasing safety at sea (91.3%), promotion of education/ training (94.6%), receiving weather reports (88.6%) and protecting local fisher-folk through surveillances with 98.6% response.</p> <p>ICT also acts to reduce vulnerability among fisher folk (72.6%) and social inclusion (50%), which makes farmers have right sense of belonging.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 8 | Education       | <p>Using ICTs, educational content can be delivered to students in their home communities thus saving travel costs. Radio and television have been used widely as educational tools since the 1920s and the 1950s, respectively in the following areas:</p> <ol style="list-style-type: none"> <li>1. Direct class teaching, where broadcast programming substitutes for teachers on a temporary basis;</li> <li>2. School broadcasting, where broadcast programming provides complementary teaching and learning resources not otherwise available; and</li> <li>3. General educational programming over community, national and international stations which provide general and informal educational opportunities.</li> </ol> <p>Teleconferencing and audio conferencing are now used extensively in education. These involve the live (real-time) exchange of voice messages over a network. Text and images such as graphs, diagrams or pictures can be exchanged along with voice messages. Non-moving visuals are added using a computer keyboard or by drawing/writing on a graphics tablet or whiteboard. Videoconferencing allows the exchange of moving images. Web-based conferencing involves the transmission of text, and graphic, audio and visual media via the Internet. Teleconferencing is used in both formal and non-formal learning contexts to facilitate teacher-learner and learner-learner discussions, as well as to access experts and other resources remotely. In open and distance learning, teleconferencing is a useful tool for providing direct instruction and learner support, minimizing learner isolation. Extensive broadband access now allows for educational content to be delivered direct to the students’ home thereby eliminating the need for student travel to distant schools when not required or when it is impractical.</p> |
| 9 | Sporting (Etal) | <p>The use of ICT in PE makes the science of sport come to life by linking both physical and mental activity. It also helps to create full-fledged students who are able to concentrate better on both practical and theoretical work. Besides, it helps students to develop a better understanding of their own body parts and that of the human body in general. It also raises the profile of P.E within the establishment by</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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|----|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                    | making the subject not only interesting, but also attractive and effective. Furthermore, it brings enthusiasm and motivation for both PE teachers and students. Information and communications technology (ICT) describe the variety of technological tools and resources used to produce, distribute, store and manage information and knowledge. In the modern era, these tools have brought revolutions to training and teaching methodologies of sports and physical education. Rapid development in ICT have introduced innovation and increases the effectiveness of training program. Using these advanced ICT tools, performance. level continues to grow and expected level of performance increases to all time high ICT improves accessibility and expand digital environment to the field of sports and physical education. It not only let a user to earn knowledge, it also helps to connect and communicate with the world. It brings changes and reforms to pedagogy of physical education and sports. Therefore, in the current study the researcher intends to discuss about some of the popularly used ICT tools in the field of sports and physical education.                                                                                                                                                                                                                                               |
| 10 | Weather monitoring | <ol style="list-style-type: none"> <li>1. In 2008, floods on the Ecuadorian coast resulted in 3,000 million hectares of non-productive land, with economic losses of around 85 million EUR. An initiative to develop an early-warning system to reduce the impact of natural disasters was implemented by a leading Latin American telecommunication company, in conjunction with two institutions involved in monitoring the impact of El Niño Phenomena in Ecuador: the National Research Centre of El Niño Phenomena and the Hydrology and Meteorology National Institute. This resulted in the development of a mobile information system of climate alerts, warning inhabitants of the Ecuadorian coastal region of climate disasters using messages sent to their mobile phones, so they can disseminate this information to their communities.</li> <li>2. Up to 5,000 automatic weather stations are to be deployed at cellular sites across Africa, where less than 300 are reporting today. This will increase accuracy of forecasts and provision of weather information via mobile phones to users and communities, including remote farmers and fishermen. The initial deployment focuses on the area around Lake Victoria in Kenya, United Republic of Tanzania and Uganda. The first 19 automatic weather stations installed have doubled the weather monitoring capacity of the Lake Victoria region.</li> </ol> |
|    | Waste Management   | ICTs can also play a role in environmental protection; waste management and environmentally friendly supply chain management. It is now possible to search websites for places to recycle ICT equipment, including mobile phones. ITU, together with the ICT industry, is developing standards for improved recycling of industrial waste, including a recommendation on communication formats                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## 7. Standardization process

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## **8. Recommendations and future use**

1. Smart phones / Tablets/ Laptops may be out of the budget of the economically poor households. Low cost devices with minimum features such as Wi-Fi, Bluetooth, cellular connectivity and long battery life are required to accelerate the use of technology in various applications in rural areas.
2. Sensors are at the bottom of pyramid for IoT systems, however, at present most of the typically used sensors are imported. Further, there is a dire need to develop new sensors all the time. Looking at these requirements, sensors need to be developed indigenously with focus on low cost, high throughput and scalable manufacturing techniques such as printing based techniques.
3. An eco-system may be created for the research & development of sensor based connected devices (IoT devices) to harness the advantages of using IoT in agriculture, fisheries, animal husbandry, health care etc. Indigenous manufacturing of such devices will make them affordable for deployment in rural areas.
4. To enhance the productivity of farmers, information generated by variety of sensors deployed in the field may be analysed by a cloud based smart agriculture app which may include crop management through smart irrigation, smart pesticide control, proactive information sharing frameworks on crops and weather (using sensors, cameras, drones etc.). Options may be explored to share this data with Call Centre for the benefits of farmers.
5. Smart City platforms should be able to manage the emergency health services to the public of city as well as rural areas by analysing data from the connected health care devices as well as respond to the calls of the public, especially in a pandemic situation.
6. ICT devices should be safe and secured. To ensure this, devices are required to be tested and certified in prescribed labs.
7. If any ICT device is hacked or exposed to any vulnerability, it should be reported. For this, a suitable mechanism is required to be created for reporting of vulnerability by manufacturer, user or service provider. All devices/gateways (to be connected directly to Public Switched Telephone Network (PSTN)/ Public Land Mobile Network (PLMN)) should have IPv6 or dual stack (IPv4 and IPv6) capability.

## **8.9. Challenges and Way forward:-**





Apart from the technology related challenges mentioned in the above figure, following challenges are also required to be addressed for the proper use of ICT solution for climate change adaptation:

1. Lack of adequate infrastructure

There is the shortage of required infrastructure like electricity, communication infrastructure for providing internet facility etc. in rural areas.

2. Illiteracy (Technical illiteracy)

There is need of literacy as well as technical literacy in India; there is a connection between education level and use of electronic means or Internet. This is a major drawback in which the users are not technically literate to use the technology. In India the literacy rate of the rural population is much less than the urban population. The government is working on creating ways to increase the literacy rate amongst rural population.

3. Affordability

In India the majority of population in rural areas depends on agriculture, animal husbandry like practices to earn their daily living. For those people, accessing Internet is a costly affair.

4. Language Dominance

The dominance of English language over the internet, limits the access of non-english speaking population. In India, most of the population speak Hindi or their regional language. Due to such irresistible domination of

English over these communication channels, computers and the internet, it becomes difficult for villagers and farmers to access these things.

5. Digital divide-

The digital divide refers to the separation that exists between individuals, communities, and businesses that have access to information and communication technology and those who do not have such access. An individual living below poverty line cannot afford a computer for himself to harness the benefits of e-governance and other online services. As the digital divide narrows, broader adoption of e-governance in the public domain becomes possible. Economic poverty is not the only cause of digital divide. It can also be caused by the lack of awareness among the people.

**9.10. References**

Appendix

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